

Spring 2010

Problem 1 (Fa80, Sp10). Let X be a compact metric space and $f : X \rightarrow X$ an isometry. Show that $f(X) = X$.

Problem 2 (Sp10). Let G be a group of order $2010 = 2 \cdot 3 \cdot 5 \cdot 67$. Determine how many subgroups of order 67 can G have, and give examples for each possible number.

Problem 3 (Sp10). Suppose that f is an entire function such that for all z , $|f(z)| \leq |z|^2$. Find all possibilities for f .

Problem 4 (Sp10). Let A and B be $n \times n$ real matrices such that $A^3 = B^5 = I_n$ and $AB = BA$. Show that $A + B$ is an invertible matrix.

Problem 5 (Sp10). Let $I = (a, b)$ be an interval containing 1 and let $y = y(t)$ be a C^∞ function on I that satisfies the differential equation

$$y' = 2^y - \frac{1}{x}$$

Prove that:

1. If $y(1) > 0$ then y is (strictly) increasing on the interval $[1, b)$.
2. The same conclusion holds if $y(1) = 0$.

Problem 6 (Sp10). How many subfields does the splitting field of $x^5 - 2$ over the rationals have?

Problem 7 (Sp81, Sp10, Sp19, Sp22). Evaluate $\int_{-\infty}^{\infty} \frac{x \sin x}{(1 + x^2)^2} dx$.

Problem 8 (Sp10). Let V be a finite dimensional vector space and $T : V \rightarrow V$ be a linear transformation. Let $W \subseteq V$ be a linear subspace such that T maps W to W . Show that the characteristic polynomial of $T|_W$, the restriction of T to W , divides the characteristic polynomial of T .

Problem 9 (Sp10). Find all solutions of the differential equation $x \frac{dy}{dx} + y = x$ for $x > 0$.

Problem 10 (Sp83, Sp10, Fa24). Let $\lambda_1, \lambda_2, \dots, \lambda_n, \dots$ be real numbers. Show that the infinite series

$$\sum_{n=1}^{\infty} \frac{e^{i\lambda_n x}}{n^2}$$

converges uniformly over $\mathbb{R}$ to a continuous limit function $f : \mathbb{R} \rightarrow \mathbb{C}$. Show, further, that the limit

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(x) \, dx$$

exists.

Problem 11 (Fa79, Fa87, Sp10, Fa15). Let $M_{n \times n}(\mathbb{F})$ be the ring of $n \times n$ matrices over a field $\mathbb{F}$. Prove that it has no 2-sided ideals except $M_{n \times n}(\mathbb{F})$ and $\{0\}$.

Problem 12 (Sp10, Sp19). How many complex numbers z are there such that $|z| < 2$, $\Im(z) > 0$, and $z^7 + e^z = 0$?

Problem 13 (Sp80, Sp10, Fa13, Sp18). Let A and B be $n \times n$ complex matrices. Prove or disprove each of the following statements:

1. If A and B are diagonalizable, so is $A + B$.
2. If A and B are diagonalizable, so is AB .
3. If $A^2 = A$, then A is diagonalizable.
4. If A is invertible and A^2 is diagonalizable, then A is diagonalizable.
5. If AB is diagonalizable and invertible then so is BA .

Problem 14 (Sp88, Sp10, Sp17). Does there exist a continuous real valued function $f(x)$, $0 \leq x \leq 1$, such that

$$\int_0^1 x f(x) \, dx = 1 \quad \text{and} \quad \int_0^1 x^n f(x) \, dx = 0$$

for $n = 0, 2, 3, 4, \dots$? Give an example or a proof that no such f exists.

Problem 15 (Sp88, Sp96, Sp10). Determine the rightmost decimal digit of $A = 17^{17^{17}}$.

Problem 16 (Sp10). Find all residues of $(\cot z)/z^2$ and use this to find the sum $1/1^2 + 1/2^2 + 1/3^2 + \dots$

Problem 17 (Su83, Sp10). Let A be an $n \times n$ Hermitian matrix satisfying the condition $A^5 + A^3 + A = 3I$. Show that $A = I$.

Problem 18 (Sp10). Prove that if all edges of the complete graph on $\binom{m+n}{m}$ points are colored either red or blue, then there is either a complete red subgraph on $m + 1$ points or a complete blue subgraph on $n + 1$ points.